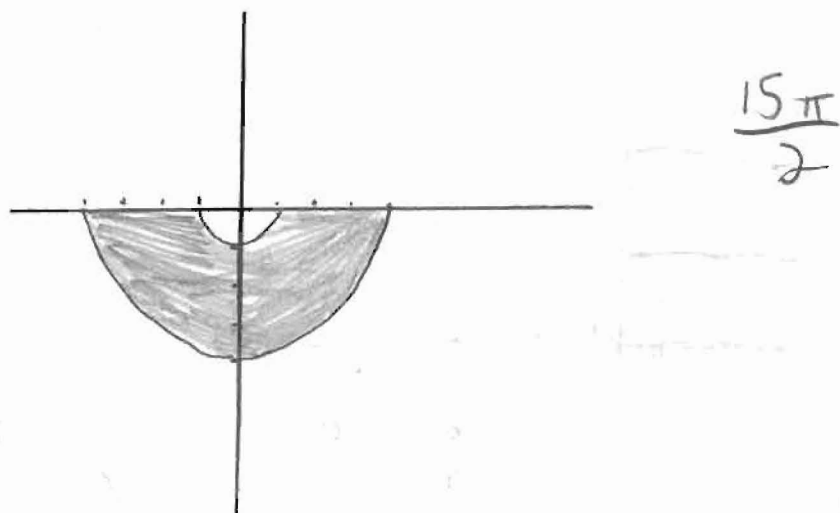


15.4 Answers

① $\int_{\pi/4}^{\pi} \int_1^2 f(r \cos \theta, r \sin \theta) r dr d\theta$

②



③ a) 0 b) $\frac{112}{3}$ c) $\frac{\pi}{2} [1 - e^{-9}]$

d) $\frac{\pi}{4} [e - 1]$

④ $\frac{\pi}{8}$

⑤ a) 8π

b) $32\pi\sqrt{3}$

c) $\frac{\pi}{3} (2 - \sqrt{2})$